

Review for Midterm 1

Dot product and hyperplanes

1.2.2, 1.3.8

Gaussian elimination

1.4.4

Rank

1.5.14, Quiz 3 Problem 2(a)

Existence/uniqueness of solutions, nonsingularity, and inverses

1.5.12, 2.3.7(a, b), Quiz 3 Problem 2(b, c)

Linear transformations

2.2.4, 2.2.7(a), Quiz 4 Problem 1

Some additional problems

These problems should require only a little computation.

Problem 1. Let

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & 3 & 0 & 4 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

- (a) One of the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ has the solution set on the left below, and the other system has the solution set on the right:

$$\mathbf{x} = s \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -4 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{x} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}.$$

Without writing anything down, can you tell which one is which?

- (b) Note that the sum of the columns of A is $\mathbf{b} = \begin{bmatrix} 3 \\ 8 \\ 1 \end{bmatrix}$. Find all solutions to $A\mathbf{x} = \mathbf{b}$.

Problem 2. Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are vectors in $\mathbb{R}^5$ such that $\mathbf{b} \in \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ if and only if

$$b_1 + 3b_4 = 0 \quad \text{and} \quad b_1 - b_2 + 2b_3 - 4b_5 = 0.$$

What is the rank of the 5×3 matrix $\begin{bmatrix} | & | & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \\ | & | & | \end{bmatrix}$?

Problem 3.

(a) Let

$$A = \begin{bmatrix} 1 & 4 & 9 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

Find all solutions to $A\mathbf{x} = \mathbf{0}$. How many solutions does $A\mathbf{x} = \begin{bmatrix} -1 \\ 7 \\ 12 \end{bmatrix}$ have?

(b) Let

$$A = \begin{bmatrix} 1 & 4 & 9 \\ 0 & 2 & 6 \end{bmatrix}$$

How many solutions does $A\mathbf{x} = \mathbf{0}$ have? Does there exist $\mathbf{b}$ such that $A\mathbf{x} = \mathbf{b}$ has no solutions? A unique solution? Infinitely many solutions?

(c) Do part (b) for the matrix

$$A = \begin{bmatrix} 1 & 4 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}.$$

Problem 4. Suppose $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is a linear transformation which satisfies

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T\left(\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad T\left(\begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Find a vector $\mathbf{v} \in \mathbb{R}^3$ such that $T(\mathbf{v}) = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$. Can you describe all such vectors $\mathbf{v}$?

Problem 5.

(a) Find x and y such that the matrices

$$\begin{bmatrix} 3 & 4 \\ 6 & x \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 5 & 4 & y \end{bmatrix}$$

are singular.

(b) Find y and z such that the matrix

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 3 \\ 5 & 4 & y & z \end{bmatrix}$$

does not have a right inverse. Does it have a left inverse?